

Matthew Minwoo Kim

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Education

Seoul National University <i>B.S. in Computer Science and Engineering</i> <i>Interdisciplinary Major in AI — Kim Jaechul AI Class</i> - GPA: 4.21/4.3 (76 credits)	Mar. 2023 – Present Seoul, South Korea
Korea Science Academy of KAIST <i>High School Diploma, Computer Science</i> - GPA: 4.14/4.3 (top 10% of the class)	Mar. 2020 – Feb. 2023 Busan, South Korea
University of Cambridge <i>International Summer School - Business and Entrepreneurship</i>	Summer 2022 Cambridge, United Kingdom

Research Experience

Vision and Learning Lab , Seoul National University <i>Undergraduate Research Intern (Advisor: Prof. Gunhee Kim)</i> Conducting research on LLM agent and multimodal AI.	Apr. 2026 – Present
Intelligent Data Systems Lab , Seoul National University <i>Undergraduate Research Opportunity Program (Advisor: Prof. Sang-goo Lee)</i> Conducted research on Text-to-SQL generation, focusing on intermediate representation and ambiguity resolution framework.	Jul. 2024 – Mar. 2025
Computer Science Research Group , Korea Science Academy of KAIST <i>Student Researcher</i> Proposed a decentralized secret sharing scheme without third-party dealers (Regeneron ISEF South Korea Finalist)	Feb. 2021 – Feb. 2022

Professional Experience

Republic of Korea Air Force <i>AI Researcher</i> - Conducted R&D on military-specific AI systems, specializing in computer vision and speech processing. - Co-authored two papers on deep learning-based infrastructure monitoring, published in the Journal of the Korean Society of Airport.	Oct. 2024 – Jul. 2026
AIBIZON AI Lab <i>Research Team Intern</i> - Conducted an empirical analysis of the RAPTOR retrieval framework. - Developed a CRDT-based system for synchronous collaboration and state management within Slack.	Mar. 2024 – Apr. 2024
Hyperithm <i>Research Team & Infra Team Intern</i> - Developed an HFT-optimized data feed to ensure ultra-low latency cryptocurrency market data ingestion. - Developed quantitative trading tools to support internal research and trading operations.	Jun. 2023 – Aug. 2023

Awards & Achievements

Army Chief of Staff Award , Military AI Contest (MAICON) 2nd place, national-level AI competition	2025
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South Korea Finalist , Regeneron ISEF(International Science and Engineering Fair)	2022
◦ Systems Software category (Team Leader)	
Grand Prize , R&E Project Presentation Contest for the Scientific Talent	2021
◦ Ministry of Science and ICT Award	

Skills

- **Languages:** Korean (Native), English (Advanced)
- **Programming Languages:** Python(PyTorch, TensorFlow), C++, Java, Typescript, Rust

Additional Information

- **Citizenship:** United States & South Korea
- **Military Service:** Serving as AI Researcher in ROK Air Force (Oct.2024–Jul.2026).

Research Interests

- Multimodal AI
- Representation Learning
- Mathematical Foundations of ML